Colleagues,

You are receiving this message as a member of a new listserv (dmon_users@whoi.edu) that is designed to facilitate communication among all of the folks using DMON2-equipped Slocum gliders. As a listserv, you are welcome to unsubscribe, or forward the listserv information at the bottom of this message to colleagues who you think should subscribe to the listserv. My hope is that we can use this listserv to keep in touch on issues or updates about the DMON2 instrument. Which brings me to the purpose of this particular email…

Over the past year or two I've collected a lot of feedback from all of you about the DMON2, both issues and suggestions for improvements. As a result, I have updated the d3host/d3read software, developed a new record-only application that allows the user to select recording parameters for themselves, developed helper software to extract/rename .wav files, and updated the documentation. This can all be found here:

https://drive.google.com/drive/folders/1xpNRwsUe2CjCpkMTZKu5AJqNnHMKv1ds?usp=sharing

In that google folder is a single zip file (dmon_2503.zip) with version 2.3 of d3host, version 2.1 of d3read, new versions of _boot.bin and _reboot.bin, the new multi-channel record-only firmware called mcrec2409.bin, a program to extract/rename .wav files, and updated instructions that includes new sections on tips/best practices and troubleshooting.

To install, simply unzip the folder and place it wherever you would like on your PC. Most of you will have already installed the DMON device drivers using the Zadig tool, so you can probably skip this. Connect a DMON2 to your PC, launch the new d3host, and follow the steps in section K of the instructions to install _reboot.bin on the DMON2. The new _reboot.bin must be installed on each of your DMON2 instruments.

I highly recommend installing and using the new versions of d3host/d3read/_boot.bin/_reboot.bin. Any firmware that you are using now will work just fine with these updates. There are two important issues that these updates address that can make for major headaches: (1) dodgy USB ports on PCs can put the DMON2 into what for lack of a better term I call "death mode" where the board is completely unresponsive and can only be revived by opening the science bay and disconnecting/reconnecting the lithium battery connector (not fun the morning of a deployment), and (2) lithium batteries becoming fully depleted when a DMON2 is not properly put into hibernation mode. The new version fixes #1, and reminds you about the correct procedure for putting a DMON2 into hibernation mode to address #2.

If you are using record-only firmware that starts with the letters "LF", you can now use mcrec2409.bin if you wish. The recording parameters in the older firmware are hard-coded, but the recording parameters for mcrec2409.bin are now user selectable, so you can select from which hydrophones to record as well as change the sampling rate and duty cycling using option b.1 in d3host (see section C in the instructions). Audio

saved in wav files are now extracted by d3read with more user-friendly names (e.g., _LF.wav instead .wav2). My thanks to Albert Miralles at SOCIB for suggesting these improvements. mcrec2409.bin is also very handy for all users to test their hydrophones prior to deployments.

Finally, I would encourage everyone who works directly with the gliders to read the “Hydrophone head” section in “Tips and best practices” of the instructions. It describes how (and how not) to handle the hydrophone head before and after it is installed in the glider. The upshot is simply this: don’t grab, twist or knock the amber dome of the hydrophone head. It is delicate and will break, and it is no fun to replace.

Please do let me know if you have any questions, concerns or issues with this update. Feel free to email via the listserv or to me directly. Please note that I will be out of the country on vacation all of next week and the early part of the following week (March 10-18), but can respond to emails thereafter.

I hope your next mission with a DMON2 is successful!

All the best,
Mark

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Update in 2503 includes the following:
(1) _boot.bin handles power transients when disconnecting DMON2 from dodgy PC USB ports
(2) _reboot.bin fixes flash read error
(3) d3host and application firmware can access non-volatile EEPROM to save recording and clock speed parameters as well as diagnostic information
(4) d3read/firmware names wav files with more user-friendly suffixes
(5) d3host reminds user of proper procedure for hibernation prior to long-term storage to address lithium battery depletion issue
(6) mcrec2409.bin allows user-selectable recording parameters and clock speed
(7) Numerous improvements in the DMON2 API for heavy (fast) processing loads